

# Emacs support for the Seed7 Programming Language

| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Keystroke                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Function                                                                                                                                                                                                                                   | Note                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                         |
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| <b>Seed7 Editing</b> <ul style="list-style-type: none"><li><a href="#">Help &amp; customization</a></li><li><a href="#">Alt File, Comments</a></li><li><a href="#">Template Expansion</a></li><li><a href="#">Seed7 abbreviations</a></li><li><a href="#">Marking</a></li><li><a href="#">Menu</a></li><li><a href="#">iMenu/ Speedbar</a></li><li><a href="#">Outline</a></li><li><a href="#">Navigation</a></li><li><a href="#">Xref Navigation</a></li><li><a href="#">Syntax-aware Indentation</a></li><li><a href="#">Compilation</a></li><li><a href="#">Seed7 Information</a></li></ul> | PEL supports for the Seed7 programming language uses <a href="#">seed7-mode</a> . PEL also provides useful <b>extensions &amp; utilities</b> that can't be implemented in the major mode.<br> The <b>seed7-mode</b> external package is installed when  the <b>pel-use-seed7</b> user-option is set to t. <ul style="list-style-type: none"><li>Seed7 files are files with .sd7 and .s7i extensions.</li></ul> The <b>seed7-mode</b> supports: <ul style="list-style-type: none"><li>Seed7 code highlighting</li><li>Insertion of Seed7 bock or line-end comments. Ability to select which type is inserted by comment-swim.<ul style="list-style-type: none"><li>PEL also provides a command to select the comment style allowing easy selection of different styles of multi-line comments, a feature provided by Emacs that PEL uses and provides an easy selection at prompt.</li></ul></li><li>Seed7 code navigation across function and procedures as well to start/end of blocks inside functions/procedure as well as enum and struct.</li><li>imenu support, allowing use of all imenu-based navigation commands and pop-up menus. Identifies callable (functions and procedures), interfaces, enums, structs.</li><li> <b>Speedbar</b> support and top menu with available commands. (see  <b>Menus</b> )</li><li>Seed7-syntax-aware auto-indentation and auto-fill-mode are supported.</li><li>Code keyword expansion to Seed7 statements with ability to jump to next field to fill with <b>tempo markers</b> and navigation to those.</li><li>outline-minor-mode to list the name of Seed7 callables. See  <b>Outline</b> for more information.</li><li>Invocation of Seed7 compiler tools to perform static analysis or compilation of Seed7 code.</li></ul> |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
| Last updated on:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-09-20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | All PEL functions that extend the <a href="#">seed7-mode</a> have names that are shown using <b>light green color</b> . See  <b>Legend</b> for more info. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
| <b>Open this PDF file.</b><br>See also:  <b>Help/Info</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div>&lt;f11&gt; SPC 7 &lt;f1&gt;</div> <div>&lt;f12&gt; &lt;f1&gt;</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ( <a href="#">pel-help-pdf</a> & optional <a href="#">OPEN-WEB-PAGE</a> )                                                                                                                                                                  | Open the  <b>L - Seed7</b> local PDF. With <b>C-u</b> , <b>M--</b> , <b>M- -1</b> or <b>C- -1</b> prefix: open the remote GitHub hosted PDF instead. If the <b>pel-flip-help-pdf-arg</b> user-option is set it's the other way around.                                                                                                                                                              |                                                                         |
|  <b>Customize</b> PEL Seed7 support                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <div>&lt;f11&gt; SPC 7 &lt;f2&gt;</div> <div>&lt;f12&gt; &lt;f2&gt;</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ( <a href="#">pel-customize-pel</a> & optional <a href="#">OTHER-WINDOW</a> )                                                                                                                                                              | Customize PEL Seed7 support. <ul style="list-style-type: none"><li>If <a href="#">OTHER-WINDOW</a> is non-nil (use <b>C-u</b>), display in another window.</li></ul>                                                                                                                                                                                                                                                                                                                 |                                                                         |
|  <b>Customize</b> Emacs Seed7 support                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <div>&lt;f11&gt; SPC 7 &lt;f3&gt;</div> <div><ul style="list-style-type: none"><li>&lt;f12&gt; &lt;f3&gt;</li><li>C-c &lt;f3&gt;</li></ul></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ( <a href="#">pel-customize-library</a> & optional <a href="#">OTHER-WINDOW</a> )                                                                                                                                                          | Customize Emacs Seed7 support: <a href="#">seed7</a> <ul style="list-style-type: none"><li>If <a href="#">OTHER-WINDOW</a> is non-nil (use <b>C-u</b>), display in another window.</li></ul>                                                                                                                                                                                                                                                                                         |                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
| <b>Show seed7-mode version information</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | C-c ?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ( <a href="#">seed7-mode-version</a> )                                                                                                                                                                                                     | Print `seed7-mode' version UTC time stamp.                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                         |
| <b>Show seed7-mode info</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"><li>C-h m</li><li>&lt;f1&gt; m</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ( <a href="#">describe-mode</a> & optional <a href="#">BUFFER</a> )                                                                                                                                                                        | Display information about the seed7-mode extracted from seed7-mode key-map and docstring. See  <b>Help/Info</b> for more on navigating Emacs help.                                                                                                                                                                                                                                                  |                                                                         |
| <b>Open file with alternate extension</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div>M-&lt;f11&gt; M-f M-f</div> <div>M-&lt;f12&gt; M-f</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ( <a href="#">pel-open-file-alternate</a> )                                                                                                                                                                                                | Open a file with same name but an alternate extension: .sd7 <--> .si7<br> The pel-file-extension-alist user option defines the extension pairs.<br>If no alternate file found, save the file basename in the kill ring and prompt for the file name to open.                                                                                                                                       |                                                                         |
| <a href="#">Comments</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | More are available to insert & manipulate comments, listed in  <b>Comments</b> . Some are duplicated here for convenience. The seed7-mode specific are listed first.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
| <b>Toggle between Seed7 (* block *) and # line style</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | C-c ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ( <a href="#">seed7-toggle-comment-style</a> & optional <a href="#">ARG</a> )                                                                                                                                                              | Toggle the Seed7 comment style between block and line comments. <ul style="list-style-type: none"><li>Optional numeric ARG, if supplied, switches to block comment style when positive, to line comment style when negative, and just toggles it when zero or left out.</li></ul> Note: the default style for all Seed7 buffers is controlled by the <b>`seed7-uses-block-comment'</b> customizable user-option. The default is line style comments.                                 |                                                                         |
| <b>Insert, realign, comment/uncomment region</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M- ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ( <a href="#">comment-dwim</a> ARG)                                                                                                                                                                                                        | Insert or realign comment on current line (or region if a region is active). <ul style="list-style-type: none"><li>On a single line, the comment is placed <i>after</i> the code. If line/region is already commented, uncomment it.</li><li><b>C-u M- ;</b> executes comment-kill</li></ul>                                                                                                                                                                                         |                                                                         |
| <b>Toggle display of comments in buffer or active region</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <f11> ; ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <a href="#">(pel-comment-dwim</a> ARG)                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Same as <b>comment-dwim</b> but comment the current line with <b>M-0 M- ;</b>                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
| <b>Change comment style for buffer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <f11> ; s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ( <a href="#">pel-comment-style</a> & optional <a href="#">CUSTOMIZE</a> )                                                                                                                                                                 | Select a comment style for the buffer: prompts with the list of available styles, showing the currently used one. Apply the choice to the current buffer. With <b>C-u</b> prefix, open the customize buffer to control selection of the default comment style for all buffers (the <b>comment-style</b> user option).                                                                                                                                                                |                                                                         |
| As of Emacs 30, Emacs supports 8 different comment styles, listed here: ➡➡                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Emacs supports several comment styles, as specified by the <b>comment-styles</b> user-option (which can be modified). Some of these styles only take effect when a region of several lines is comments. By changing the style you can create the boxed comments, for instance and also uncomment the box comment with comment-swim (bound to <b>M- ;</b> ) and then change for another comment style in the same buffer. <ul style="list-style-type: none"><li>The style selected by the command only affects the current buffer. It is not persistent. The persistent setting is the <b>comment-style</b> user option.<ul style="list-style-type: none"><li>0 = plain: Start in column 0 (do not indent), as in Emacs-20</li><li>1 = indent-or-triple: Start in column 0, but only for single-char starters</li><li>2 = indent: Full comment per line, ends not aligned</li><li>3 = aligned: Full comment per line, ends aligned</li><li>4 = box: Full comment per line, ends aligned, + top and bottom</li><li>5 = extra-line: One comment for all lines, end on a line by itself</li><li>6 = multi-line: One comment for all lines, end on last commented line</li><li>7 = box-multi: One comment for all lines, + top and bottom</li></ul></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
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| <b>Seed7 Code Template insertion &amp; expansion</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | The <a href="#">seed7-mode</a> supports a set of code keyword expansion to Seed7 statements with ability to jump to next field to fill with <b>tempo markers</b> and navigation to these fields to complete the template easily. <ul style="list-style-type: none"><li>Code keyword expansion is performed by the <b>seed7-complete-statement-or-indent</b> command, bound to the <b>&lt;tab&gt;</b> key.</li><li>To use keyword expansion: type the keyword then type <b>&lt;tab&gt;</b> to expand the keyword into the corresponding code that will be properly indented.</li></ul> There are 2 groups of supported keywords. <ul style="list-style-type: none"><li>The keywords shown in the first part of the table expand to their corresponding code template when the keyword is the only word on the line and point is placed just after the last keyword character.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
| <b>Top level or block declarations.</b><br><br>Type the keyword at the beginning of the line and hit <b>&lt;tab&gt;</b> to expand the corresponding code.                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>inc</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | include statement                                                                                                                                                                                                                          | <b>for</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <a href="#">for statement</a>                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>const</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <a href="#">constant declaration</a>                                                                                                                                                                                                       | <b>foru</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <a href="#">for-until statement</a>                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>var</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <a href="#">variable declaration</a>                                                                                                                                                                                                       | <b>fors</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <a href="#">for-step statement</a>                                      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>proc</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">procedure declaration</a>                                                                                                                                                                                                      | <b>fore</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <a href="#">for-each statement</a>                                      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>func</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">function declaration</a>                                                                                                                                                                                                       | <b>foreu</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">for-each statement</a> combined with an until condition     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>funcs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <a href="#">short function declaration</a>                                                                                                                                                                                                 | <b>forek</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">for-each-key statement</a>                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>enum</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">enum type declaration</a>                                                                                                                                                                                                      | <b>foreku</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <a href="#">for-each-key statement</a> combined with an until condition |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>struct</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <a href="#">struct type declaration</a>                                                                                                                                                                                                    | <b>fork</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <a href="#">for-key statement</a>                                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>case</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">case statement</a>                                                                                                                                                                                                             | <b>forku</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">for-key statement</a> combined with an until condition      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>if</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <a href="#">if statement</a>                                                                                                                                                                                                               | <b>repeat</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <a href="#">repeat - until statement</a>                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ife</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <a href="#">if statement</a> with an else clause                                                                                                                                                                                           | <b>while</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">while statement</a>                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ifei</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">if statement</a> with an elsif clause                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ifeie</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <a href="#">if statement</a> with an elsif and an else clause                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
| <b>Parameter declarations</b><br>Also expand with <b>&lt;tab&gt;</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"><li>The second group of keywords are expanded when the keyword precedes a closing parenthesis; they are use to expand the parameter declarations.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>in</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Declaration of an <a href="#">in-parameter</a> .                                                                                                                                                                                           | <b>callbn</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Declaration of a <a href="#">call-by-name parameter</a> .               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>inout</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Declaration of an <a href="#">inout-parameter</a> .                                                                                                                                                                                        | <b>ref</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Declaration of a <a href="#">reference-parameter</a> .                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>invar</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Declaration of an <a href="#">in-var-parameter</a> .                                                                                                                                                                                       | <b>val</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Declaration of a <a href="#">value-parameter</a> .                      |
| <b>Expand keyword or indent</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>&lt;tab&gt;</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ( <a href="#">seed7-complete-statement-or-indent</a> )                                                                                                                                                                                     | If point follows a valid code keyword properly located, this perform code expansion, leaving point at the first location that must be filled. <ul style="list-style-type: none"><li>In that case you can then type &lt;backtab&gt; to move to the next field that needs to be filled (or has already been filled). Those are <b>tempo markers</b> that stay in the buffer until the buffer is closed.</li></ul> If point is located anywhere else indent the line or selected block. |                                                                         |
| <b>Move to next field</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>&lt;backtab&gt;</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ( <a href="#">tempo-forward-mark</a> )                                                                                                                                                                                                     | Move point to the next <b>tempo marker</b> , the next template field to fill.                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                         |

| Description                                                                                       | Keystroke                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                    | Function              |                                                                                                                                                                                                                                                                                                              | Note                         |                  |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------|
| <b>Seed7-specific abbreviations</b>                                                               | The <b>seed7-mode</b> supports Seed7-specific abbreviations for <b>Emacs abbrev-mode</b> when the <b>seed7-support-abbrev-mode</b> customizable user-option is on (the default).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| See also:<br><a href="#">⌘ Abbreviations</a>                                                      | <ul style="list-style-type: none"><li>All abbreviation and their text expansion are set by the <b>seed7-abbreviations</b> customizable user-option list.<ul style="list-style-type: none"><li>The default list is shown below. All abbreviations start with a semi-colon.</li><li>You can modify the default and add other abbreviations through customization.</li></ul></li><li>These abbreviations are <i>system</i> abbreviations, treated specially by the abbrev-mode in the sense that youcanngt modify them dynamically via the abbrev-mode commands. But you don't need to since they can be modified by customization.</li><li>Of course you can create other abbreviations that help you write code or comments. See <a href="#">⌘ Abbreviations</a> for more details related to abbreviations in Emacs.</li><li>To expand the abbreviations, the <b>abbrev-mode</b> must be active: type the abbreviation followed by a word-separating character, such as <b>&lt;space&gt;</b>, <b>&lt;RET&gt;</b>, semi-colon, period, comma, etc..</li></ul> |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
|                                                                                                   | 👉 Use the <b>list-abbrevs</b> command to list all abbreviations ( <b>&lt;f11&gt; a M-l</b> with PEL), including the following Seed7-specific ones. The list are shown in sorted order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| <b>Pragmas &amp; in-statement keywords</b>                                                        | pragmas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                    | in-statement keywords |                                                                                                                                                                                                                                                                                                              | in-middle statement keywords |                  |
|                                                                                                   | <b>;de</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | decls                                              | <b>;fo</b>            | forward                                                                                                                                                                                                                                                                                                      | <b>;dt</b>                   | downto           |
|                                                                                                   | <b>;in</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | info                                               | <b>;n</b>             | new                                                                                                                                                                                                                                                                                                          | <b>;exc</b>                  | exception        |
|                                                                                                   | <b>;li</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | library                                            | <b>;no</b>            | noop                                                                                                                                                                                                                                                                                                         | <b>;lo</b>                   | local            |
|                                                                                                   | <b>;msg</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | message                                            | <b>;ra</b>            | raise                                                                                                                                                                                                                                                                                                        | <b>;pa</b>                   | param            |
|                                                                                                   | <b>;na</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | names                                              | <b>;rt</b>            | return                                                                                                                                                                                                                                                                                                       | <b>;rg</b>                   | range            |
|                                                                                                   | <b>;syn</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | syntax                                             |                       |                                                                                                                                                                                                                                                                                                              | <b>;rs</b>                   | result           |
|                                                                                                   | <b>;sys</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | system                                             | <b>;tr</b>            | trace                                                                                                                                                                                                                                                                                                        | <b>;st</b>                   | step             |
| <b>Block clause keywords</b>                                                                      | block clause keywords                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
|                                                                                                   | <b>;ct</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | catch                                              | <b>;e</b>             | else                                                                                                                                                                                                                                                                                                         | <b>;o</b>                    | otherwise        |
|                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                    | <b>;ei</b>            | elsif                                                                                                                                                                                                                                                                                                        | <b>;w</b>                    | when             |
| <b>Pre-defined types</b>                                                                          | pre-defined types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
|                                                                                                   | <b>;a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | array                                              | <b>;db</b>            | database                                                                                                                                                                                                                                                                                                     | <b>;rat</b>                  | rational         |
|                                                                                                   | <b>;bi</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | bigInteger                                         | <b>;du</b>            | duration                                                                                                                                                                                                                                                                                                     | <b>;rf</b>                   | reference        |
|                                                                                                   | <b>;br</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | bigRational                                        | <b>;en</b>            | enum                                                                                                                                                                                                                                                                                                         | <b>;rfl</b>                  | ref_list         |
|                                                                                                   | <b>;b3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | bin32                                              | <b>;ex</b>            | expr                                                                                                                                                                                                                                                                                                         | <b>;s</b>                    | set              |
|                                                                                                   | <b>;b6</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | bin64                                              | <b>;fi</b>            | file                                                                                                                                                                                                                                                                                                         | <b>;sq</b>                   | sqlStatement     |
|                                                                                                   | <b>;bt</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | bitset                                             | <b>;fs</b>            | fileSys                                                                                                                                                                                                                                                                                                      | <b>;sti</b>                  | string           |
|                                                                                                   | <b>;bo</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | boolean                                            | <b>;fl</b>            | float                                                                                                                                                                                                                                                                                                        | <b>;stu</b>                  | struct           |
|                                                                                                   | <b>;bs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | bstring                                            | <b>;h</b>             | hash                                                                                                                                                                                                                                                                                                         | <b>;tx</b>                   | text             |
|                                                                                                   | <b>;ca</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | category                                           | <b>;i</b>             | integer                                                                                                                                                                                                                                                                                                      | <b>;ti</b>                   | time             |
|                                                                                                   | <b>;c</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | char                                               | <b>;ob</b>            | object                                                                                                                                                                                                                                                                                                       | <b>;ty</b>                   | type             |
|                                                                                                   | <b>;cf</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | clib_file                                          | <b>;pro</b>           | process                                                                                                                                                                                                                                                                                                      | <b>;v</b>                    | void             |
|                                                                                                   | <b>;co</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | color                                              | <b>;pr</b>            | program                                                                                                                                                                                                                                                                                                      | <b>;pw</b>                   | PRIMITIVE_WINDOW |
|                                                                                                   | <b>;cx</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | complex                                            |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| <b>Pre-defined constants</b>                                                                      | pre-defined constants                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
|                                                                                                   | <b>;em</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | empty                                              | <b>;f</b>             | FALSE                                                                                                                                                                                                                                                                                                        | <b>;inf</b>                  | Infinity         |
|                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                    | <b>;t</b>             | TRUE                                                                                                                                                                                                                                                                                                         |                              |                  |
| <b>Pre-defined variables</b>                                                                      | pre-defined variables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
|                                                                                                   | <b>;ck</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CONSOLE_KEYBOARD                                   | <b>;sc</b>            | STD_CONSOLE                                                                                                                                                                                                                                                                                                  | <b>;sn</b>                   | STD_NULL         |
|                                                                                                   | <b>;gk</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | GRAPH_KEYBOARD                                     | <b>;se</b>            | STD_ERR                                                                                                                                                                                                                                                                                                      | <b>;so</b>                   | STD_OUT          |
|                                                                                                   | <b>;kb</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | KEYBOARD                                           | <b>;si</b>            | STD_IN                                                                                                                                                                                                                                                                                                       |                              |                  |
| <b>Errinfo values</b>                                                                             | errinfo values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
|                                                                                                   | <b>;ok</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | OKAY_NO_ERROR                                      | <b>;dse</b>           | DESTROY_ERROR                                                                                                                                                                                                                                                                                                | <b>;me</b>                   | MEMORY_ERROR     |
|                                                                                                   | <b>;ae</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ACTION_ERROR                                       | <b>;fe</b>            | FILE_ERROR                                                                                                                                                                                                                                                                                                   | <b>;ne</b>                   | NUMERIC_ERROR    |
|                                                                                                   | <b>;ce</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | COPY_ERROR                                         | <b>;ge</b>            | GRAPHIC_ERROR                                                                                                                                                                                                                                                                                                | <b>;oe</b>                   | OVERFLOW_ERROR   |
|                                                                                                   | <b>;cre</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CREATE_ERROR                                       | <b>;ie</b>            | INDEX_ERROR                                                                                                                                                                                                                                                                                                  | <b>;re</b>                   | RANGE_ERROR      |
|                                                                                                   | <b>;dbe</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DATABASE_ERROR                                     | <b>;ine</b>           | IN_ERROR                                                                                                                                                                                                                                                                                                     |                              |                  |
| <b>Marking</b>                                                                                    | The <b>seed7-mode</b> support specialized marking. It is also compatible with other Emancs native and package commands. See <a href="#">⌘ Marking</a> for more information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| <b>Mark current callable</b>                                                                      | <b>C-M-h</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <a href="#">(seed7-mark-defun)</a>                 |                       | <div>Mark the current Seed7 function or procedure.<ul style="list-style-type: none"><li>Put the mark at the end and point at the beginning.</li><li>If point is before or between 2 functions or procedure, mark the next one.</li></ul></div>                                                               |                              |                  |
| <b>Menu Control</b>                                                                               | The <b>seed7-mode</b> supports the top <a href="#">⌘ Menus</a> for more information on how to activate the Emacs menu. Seed7 specific information is in the <b>Seed7</b> menu section.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| <b>iMenu Control</b>                                                                              | The <b>seed7-mode</b> supports <a href="#">⌘iMenu</a> symbol identification and <a href="#">⌘ Speedbar</a> facility. All of <a href="#">⌘iMenu</a> and <a href="#">⌘ Speedbar</a> commands are available for Seed7 code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| <b>Toggle iMenu grouping of callable: function &amp; procedures listed separately or together</b> | <b>C-c %</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <a href="#">(seed7-toggle-menu-callable-list)</a>  |                       | <div>Change the way callables are listed inside the current buffer menu.<ul style="list-style-type: none"><li>Toggles listing them together or separately.</li><li>When listed separately the function and procedures are listed inside their own group, otherwise they are listed together.</li></ul></div> |                              |                  |
| <b>Toggle iMenu sorting</b>                                                                       | <b>C-c =</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <a href="#">(seed7-toggle-menu-sorting)</a>        |                       | Toggle displaying menu entries in code order or sorted order.                                                                                                                                                                                                                                                |                              |                  |
| <b>Seed7 Outlining</b>                                                                            | The <b>seed7-mode</b> supports Emacs outline minor mode. Some info is shown below See <a href="#">⌘ Outline</a> for the complete list.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| <b>Outline minor mode</b>                                                                         | <b>&lt;f11&gt; M-l</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <a href="#">(outline-minor-mode</a> &optional ARG) |                       | Toggle Outline minor mode. Headings are defined by outline variables.                                                                                                                                                                                                                                        |                              |                  |
| <b>Hide all bodies</b>                                                                            | <b>C-c C-t</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <a href="#">(outline-hide-body)</a>                |                       | Collapse the body of all blocks, leaving the first line visible.                                                                                                                                                                                                                                             |                              |                  |
|                                                                                                   | <div><div>• <b>&lt;f2&gt; t</b></div><div>• <b>C-c @ C-t</b></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| <b>Show all</b>                                                                                   | <b>C-c C-a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <a href="#">(outline-show-all)</a>                 |                       | Show all: expand the body of all blocks.                                                                                                                                                                                                                                                                     |                              |                  |
|                                                                                                   | <div><div>• <b>&lt;f2&gt; a</b></div><div>• <b>C-c @ C-a</b></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| <b>Show subtree</b>                                                                               | <b>C-c C-s</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <a href="#">(outline-show-subtree)</a>             |                       | Show everything after this heading at deeper levels.                                                                                                                                                                                                                                                         |                              |                  |
|                                                                                                   | <div><div>• <b>&lt;f2&gt; s</b></div><div>• <b>C-c @ C-s</b></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |
| <b>Hide others</b>                                                                                | <b>C-c C-o</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <a href="#">(outline-hide-other)</a>               |                       | Collapse everything except the current block.                                                                                                                                                                                                                                                                |                              |                  |
|                                                                                                   | <div><div>• <b>&lt;f2&gt; o</b></div><div>• <b>C-c @ C-o</b></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                       |                                                                                                                                                                                                                                                                                                              |                              |                  |

| Description                                                                                                                                                                                                                                                                                                                                                                              | Keystroke                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Function                                                                | Note                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <u>Code Navigation</u>                                                                                                                                                                                                                                                                                                                                                                   | The <b>seed7-mode</b> supports syntax-aware procedure/function as well as block aware navigation commands <ul style="list-style-type: none"><li>• PEL provides some extra key bindings to Emacs native navigation commands.</li><li>• The <b>seed7-mode</b> also supports imenu-compliant parsing which enables the ability to use a large set of navigation packages.<ul style="list-style-type: none"><li>• See <a href="#">navigation by symbol definition</a> in the <a href="#">Navigation</a> page for more information.</li></ul></li><li>• The <b>seed7-mode</b> navigation commands display the name and type of block found when the <b>seed7-verbose-navigation</b> user-option is turned on (set to <b>t</b>).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <u>Shift-Selection</u>                                                                                                                                                                                                                                                                                                                                                                   | If you press and hold the <b>shift</b> key while typing a movement command, that sets the mark before moving point (Emacs name for cursor) so that the region extends from the original point to its new position. This is called: <b>Shift-Selection</b> . <ul style="list-style-type: none"><li>• <b>Shift selection</b> is supported by some navigation commands, not all. The following symbols are used to identify whether the command supports shifts selection:<ul style="list-style-type: none"><li>•  This command supports shift selection in GUI and terminal mode.</li><li>•  This command supports shift selection only in GUI mode.</li><li>•  This command supports shift selection in GUI mode and also in terminal mode under some conditions (described in the description cell for the command).</li><li>•  This command does <b>not</b> support shift selection. Sometimes for this you can first set the mark before moving.</li></ul></li><li>• Pressing the Shift key when using the key binding for commands that do not show any of these 3 arrows have no impact on the shift selection (and may be inappropriate for the command).</li></ul> |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <u>Move Point</u>                                                                                                                                                                                                                                                                                                                                                                        | The following sub-sections describe how to navigate across various types of textual and syntactical entities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <ul style="list-style-type: none"><li>• <u>by sexp</u></li></ul>                                                                                                                                                                                                                                                                                                                         | seed7-mode modifies the way Emacs standard sexp navigation command works; it allows navigation across <b>any</b> Seed7 block construct.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>block forward</b><br><br><br><br><br>                                                                                              | <ul style="list-style-type: none"><li>• <b>C-M-&lt;right&gt;</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | (forward-sexp &optional N)                                              | Move forward to the end of: <ul style="list-style-type: none"><li>• Seed7 ( * ... * ) nested block comments,</li><li>• single and consecutive # line-end comments,</li><li>• nested statement blocks,</li><li>• function and procedure blocks.</li><li>• const/var array and const/var set definition block end lines,</li><li>• matching paren for ( , [ or { .</li><li>• With argument N, do it N times. Negative N means move backward movement.</li></ul>  <b>C-M-&lt;right&gt;</b> does not work on Windows, but <b>H-&lt;right&gt;</b> does.                                                                                                                          |  |
|                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"><li>• <b>Esc C-&lt;right&gt;</b> </li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"><li>• <b>C-M-f</b></li><li>• <b>C-[ C-f</b></li><li>• <b>Esc C-f</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|  With PEL: if you want to use <b>Esc C-&lt;right&gt;</b> binding you must ensure that <b>pel-windmove-on-esc-cursor</b> user option is set to nil.                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|  Several Linux distros map <b>C-M-&lt;right&gt;</b> to desktop workspace operation. <ul style="list-style-type: none"><li>• You can either use another key binding or change Linux key binding in Systems-&gt;settings-&gt;keyboard-&gt;shortcuts to prevent it from using that key sequence.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>Up/inside sexp hierarchy</b><br><br><br><br>                                                                                       | <ul style="list-style-type: none"><li>• <b>C-M-&lt;up&gt;</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | (backward-up-list &optional ARG ESCAPE-STRINGS NO-SYNTAX-CROSSING)      | Move backward out to the beginning of: <ul style="list-style-type: none"><li>• Seed7 ( * ... * ) nested block comments,</li><li>• single and consecutive # line-end comments,</li><li>• nested statement blocks,</li><li>• function and procedure blocks.</li><li>• const/var array and const/var set definition block start lines,</li><li>• matching paren for ) , ] or } .</li><li>• With argument N, do it N times. Negative N means move forward movement.</li></ul>  <b>C-M-&lt;up&gt;</b> does not work on Windows, but <b>H-&lt;up&gt;</b> does.                                                                                                                  |  |
|                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"><li>• <b>Esc C-&lt;up&gt;</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"><li>• <b>C-M-u</b></li><li>• <b>C-[ C-u</b></li><li>• <b>Esc C-u</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|  With PEL: if you want to use <b>Esc C-&lt;up&gt;</b> binding you must ensure that <b>pel-windmove-on-esc-cursor</b> user option is set to nil.                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <ul style="list-style-type: none"><li>• <u>by defun</u></li></ul>                                                                                                                                                                                                                                                                                                                        | The commands move point by Seed7 function and procedure definitions.<br> In PEL: <ul style="list-style-type: none"><li>• The <b>&lt;f12&gt;</b> cursor key mappings use <b>&lt;up&gt;</b> and <b>&lt;down&gt;</b> to move to the beginning or end of the function, procedure or other blocks.</li><li>• The <b>&lt;f6&gt;</b> cursor key mapping use <b>&lt;up&gt;</b> and <b>&lt;down&gt;</b> to move to the beginning or end of the function or procedure.</li><li>• The <b>&lt;f6&gt;</b> cursor key mapping use <b>&lt;right&gt;</b> and <b>&lt;left&gt;</b> to move to the beginning or end of the next/previous function or procedure.</li></ul> <ul style="list-style-type: none"><li>• The advantage of the <b>&lt;f6&gt;</b> and <b>&lt;f12&gt;</b> key bindings is they support <b>Shift-Selection</b> for Emacs in terminal mode, as opposed to the key bindings that sue the Control key which can only support <b>Shift-Selection</b> when Emacs is running in Graphics mode.</li><li>• After moving successfully, unless you did not push the mark, you can go back to previous location with: <b>M-` , &lt;f6&gt; &lt;f6&gt;</b> or <b>&lt;f11&gt; . `</b></li></ul>                                                                                                                                                                                                                                       |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>Backward to beginning of defun</b><br><br><br>                                                                                                                                                                  | <ul style="list-style-type: none"><li>• <b>&lt;f6&gt; &lt;up&gt;</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (seed7-beg-of-defun &optional N SILENT DONT-PUSH-MARK)                  | Move backward to the beginning of a defun. <ul style="list-style-type: none"><li>• With ARG, do it that many times. Negative ARG means move forward to the ARGth following beginning of defun.</li><li>• Prints the name of the function or procedure in the message area.</li><li>• Supports <b>Shift-Selection</b> in graphics mode. <b>&lt;f6&gt;&lt;up&gt;</b> supports it in terminal mode too.</li></ul>                                                                                                                                                                                                                                                                                                                                               |  |
|                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"><li>• <b>C-M-a</b></li><li>• <b>C-M-&lt;home&gt;</b></li><li>• <b>C-[ C-a</b></li><li>• <b>Esc C-a</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>Forward to end of defun</b><br><br><br>                                                                                                                                                                         | <ul style="list-style-type: none"><li>• <b>&lt;f6&gt; &lt;down&gt;</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | (seed7-end-of-defun &optional N SILENT DONT-PUSH-MARK)                  | Move forward to next end of defun. <ul style="list-style-type: none"><li>• With argument, do it that many times. Negative argument -N means move back to Nth preceding end of defun.</li><li>• Prints the name of the function or procedure in the message area.</li><li>• Supports <b>Shift-Selection</b> in graphics mode. <b>&lt;f6&gt;&lt;down&gt;</b> supports it in terminal mode too.</li></ul>                                                                                                                                                                                                                                                                                                                                                       |  |
|                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"><li>• <b>C-M-e</b></li><li>• <b>C-M-&lt;end&gt;</b></li><li>• <b>C-[ C-e</b></li><li>• <b>Esc C-e</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>Move Forward to beginning of next defun</b><br>                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"><li>• <b>C-c C-n</b></li><li>• <b>&lt;f6&gt; &lt;right&gt;</b></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | (seed7-beg-of-next-defun &optional N SILENT DONT-PUSH-MARK)             | Move forward to the beginning of the next function or procedure. <ul style="list-style-type: none"><li>• With optional argument N, repeat the search that many times and succeed only when that many function or procedures are found.<ul style="list-style-type: none"><li>• A value of zero means no action. A negative value is not allowed and raises a user error.</li></ul></li><li>• Unless SILENT, the function prints a message showing the name of the found function or procedure.</li><li>• When a new function or procedure is found the function pushes the mark unless DONT-PUSH-MARK is non-nil. Pushing the mark allows future pop to go back to the original position with <b>C-u C-SPC</b>.</li><li>• Supports shift selection.</li></ul> |  |
| <b>Backward to end of previous define</b><br><br> will be replaced                                                                                                                                                 | <b>&lt;f6&gt; &lt;left&gt;</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (pel-end-of-previous-defun &optional SILENT DONT-PUSH_MARK)             | Move backwards to the end of the previous function definition. <ul style="list-style-type: none"><li>• Issue user error not find end of previous function unless SILENT is non-nil.</li><li>• If the end of previous function is found, push the start location to the mark ring unless DONT-PUSH_MARK is non-nil.</li><li>• Supports <b>Shift-Selection</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                      |  |
| <b>Forward to end of current block statement</b><br>                                                                                                                                                                                                                                                  | <b>&lt;f12&gt; &lt;down&gt;</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | (seed7-to-block-forward)                                                | Move forward from the beginning of a Seed7 block to its end. <ul style="list-style-type: none"><li>• Supports the Seed7 if/end if, block/end block, case/end case, enum/end enum, for/end for, repeat/until, struct/end struct, while/end while. It also supports moving to the end of a function or a procedure.</li><li>• Supports <b>Shift-Selection</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                       |  |
| <b>Backward to beginning of current block statement</b><br>                                                                                                                                                                                                                                           | <b>&lt;f12&gt; &lt;up&gt;</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (seed7-to-block-backward)                                               | Move backward from the end of a Seed7 block to its beginning. <ul style="list-style-type: none"><li>• supports the Seed7: if/end if, block/end block, case/end case, enum/end enum, for/end for, repeat/until, struct/end struct, while/end while. It also supports moving to the end of a function or a procedure.</li><li>• Supports <b>Shift-Selection</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                     |  |
| <b>Move to top of block</b><br>                                                                                                                                                                                                                                                                       | <b>C-c C-t</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (seed7-to-top-of-block)                                                 | Move point to the top of the current block; out of any nesting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>Move to block backward</b><br>                                                                                                                                                                                                                                                                     | <b>C-c C-a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (seed7-to-block-backward &optional AT-BEGINNING-OF-LINE DONT-PUSH-MARK) | Move backward from block end to its beginning. <ul style="list-style-type: none"><li>• Move point to the beginning of the block keyword or comment.</li><li>• If point moves to the indenting area as a result, and AT-BEGINNING-OF-LINE optional argument is set, move point to the beginning of the line.</li><li>• Push mark unless DONT-PUSH-MARK is non-nil. Supports shift-marking.</li><li>• Return found position if found, nil if nothing found.</li></ul>                                                                                                                                                                                                                                                                                          |  |
| <b>Move to block forward</b><br>                                                                                                                                                                                                                                                                      | <b>C-c C-e</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (seed7-to-block-forward &optional DONT-PUSH-MARK)                       | Move forward from the block beginning to its end. <ul style="list-style-type: none"><li>• Handle function and forward declarations blocks.</li><li>• Push mark unless DONT-PUSH-MARK is non-nil. Supports shift-marking.</li><li>• Return found position or nil if nothing found.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |

| Description                                                                                                                                                                                     | Keystroke                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Function                             | Note                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <div>Cross Reference Navigation</div> <div> See also: <a href="#">↗ Xref</a> </div> <div>  </div> </div> | <div> The Seed7 programming language can inspect itself: it is reflective in the sense that it can be parsed easily by the simple Seed7 code. <ul style="list-style-type: none"> <li>The <a href="#">seed7-mode</a> takes advantage of this to provide quick cross reference navigation without the need of a language server or ctags parsing.</li> <li>The <a href="#">seed7-mode</a> provides the <a href="#">s7xref.sd7</a> Seed7 program that acts as a cross reference parser and is used as the backend for Emacs xref. <ul style="list-style-type: none"> <li>This program can be interpreted (the most flexible option) or compiled.</li> <li>Identify the location of the program in the seed7-xref customizable user-option in the seed7 customization buffer (access it via <b>C-c C</b> or <b>&lt;f12&gt; &lt;f3&gt;</b>.</li> </ul> </li> <li>Once the <a href="#">s7xref.sd7</a> program properly in place, you can use the following commands to navigate quickly across Seed7 code.</li> <li>Note that the <a href="#">seed7-mode</a> back-end implementation of xref-find-definition does more than only looking up for identifiers managed by the <a href="#">s7xref.sd7</a> program: it is also aware of block and file scope and can identify the location of a local or global variable.</li> <li>Program used identified by <a href="#">seed7-xref</a> user-option. The default should identify the executable file with absolute path. Modify if necessary.</li> </ul> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Find definition of identifier at point                                                                                                                                                          | <b>M- .</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | (xref-find-definitions IDENTIFIER)   | <div> Grab symbol at point and move cursor to its definition. <ul style="list-style-type: none"> <li>If there are more than one match, prompt in the *xref* buffer.</li> <li>To search for a symbol entered manually, type <b>C-u M- .</b></li> <li>With dumb-jump this performs a search using ag, ripgrep or git grep if available.</li> </ul> </div>                                                                                                                                             |
| Go back to where M-. was last issued                                                                                                                                                            | <b>M- ,</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | (xref-pop-marker-stack)              | <ul style="list-style-type: none"> <li>Pop back to where M-. was last invoked.</li> <li>Marker depth is controlled by the <b>xref-marker-ring-length</b> user option.</li> </ul>                                                                                                                                                                                                                                                                                                                    |
| <div> <div>Syntax-aware automatic Indentation</div> <div> See also <a href="#">↗ Indentation</a> </div> </div>                                                                                  | <div> Emacs approach to indentation control is based on what the major-mode provides. For Seed7 code, the indentation is controlled by logic implemented by the <a href="#">seed7-mode</a>. Unless explicitly disabled by setting the <b>seed7-auto-indent</b> user-option to nil, the <b>&lt;tab&gt;</b> and <b>&lt;return&gt;</b> key perform syntax-aware automatic indentation of Seed7 code. The <b>&lt;return&gt;</b> key also supports the auto-fill-mode. <ul style="list-style-type: none"> <li>When you type code and hit the <b>&lt;return&gt;</b> key the line you just typed is indented to the location corresponding to the Seed7 indentation rules.</li> <li> When you press the <b>&lt;tab&gt;</b> key <i>anywhere</i> on the line, the <a href="#">seed7-mode</a> code shakes if the current line is properly indented and updates the indentation if required. If it's already properly indented, then nothing occurs! <ul style="list-style-type: none"> <li>That may seem unusual behaviour for new Emacs users. But that's the way Emacs deals with indentation in all modes!! It is the expected behaviour for Emacs.</li> <li>If you want to insert spaces or hard tab character, you have to use another key combination. See <a href="#">↗ Indentation</a> for more information on this topic.</li> </ul> </li> <li>Note that in Seed7 buffers, the <b>&lt;tab&gt;</b> key is bound to the <b>seed7-complete-statement-or-indent</b> command which check if point (the cursor location) follows a code keyword that should be expanded and if so performs the expansion instead of the indentation. Type <b>&lt;tab&gt;</b> again to perform the indentation in that case.</li> <li>The number of columns used for each indentation level is controlled by the <b>seed7-indent-width</b> user-option, which defaults to 2.</li> <li>Emacs can use hard tabs as appropriate when you activate the <b>indent-tabs-mode</b>. If it is off Emacs only uses space characters.</li> <li>The <a href="#">seed7-mode</a> also supports standard Emacs indentation commands.</li> </ul> </div> |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Auto-fill-mode                                                                                                                                                                                  | The <a href="#">seed7-mode</a> supports Emacs auto-fill-mode, useful when typing comments. See the <a href="#">↗ Fill/Justify</a> page and the <b>pel-comment-style</b> command above.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expand keyword or indent                                                                                                                                                                        | <b>&lt;tab&gt;</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (seed7-complete-statement-or-indent) | <div> If point follows a valid code keyword properly located, this perform code expansion, leaving point at the first location that must be filled. <ul style="list-style-type: none"> <li>In that case you can then type &lt;backtab&gt; to move to the next field that needs to be filled (or has already been filled). Those are <b>tempo markers</b> that stay in the buffer until the buffer is closed.</li> </ul> If point is located anywhere else indent the line or selected block. </div> |
| Indent enclosing block                                                                                                                                                                          | <b>C-M-q</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | (seed7-indent-block)                 | Indent the block enclosing point. Do not move point.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Refill/justify                                                                                                                                                                                  | <b>M-q</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (seed7-fill)                         | Refill/justify comment and string paragraph, re-indent current code block.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <div> <div>Compilation</div> <div>  </div> </div>                                                            | <div> The Seed7 source code is either interpreted or compiled. In both cases you can verify it's validity by performing a static check of the code, an operation that does not generate any binary file but perform the same language checking that the compiler will do. <ul style="list-style-type: none"> <li><b>seed7-checker</b> : Identifies the Seed7 static checker program. Defaults to <b>s7check</b>, without path. You can identify the path of the executable file if needed.</li> <li><b>seed7-compiler</b> : Identifies the Seed7 static checker program. Defaults to <b>s7c</b>, without path. You can identify the path of the executable file if needed.</li> </ul> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <div> <div>Static check or compile Seed7 file</div> <div> See <a href="#">↗ Compilation Mode</a> </div> <div> static check: <b>C-c C-c</b><br/> compile: <b>C-u C-c C-c</b> </div> </div>       | <div> <div><b>C-c C-c</b></div> <div><b>&lt;f12&gt; c</b></div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | (seed7-compile &optional COMPILE)    | <div> Static check current Seed7 file, show errors in compilation-mode buffer. <ul style="list-style-type: none"> <li>With optional COMPILE argument: compile the file to executable instead. <ul style="list-style-type: none"> <li>Any argument&gt;. Use <b>C-u C-c C-c</b> or <b>M-0 M-&lt;F12&gt; c</b></li> </ul> </li> </ul> </div>                                                                                                                                                           |
|                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>For example: type <b>C-u &lt;f12&gt; c</b> for compiling the file. Without the <b>C-u</b> prefix it just static checks the file, an operation that is much faster.</li> <li>The static analysis is performed by the command identified by the <b>seed7-checker</b> user-option, which defaults to s7-check. <ul style="list-style-type: none"> <li>You can specify any command with or without its path.</li> </ul> </li> <li>The compilation is performed by the command identified by the <b>seed7-compiler</b> user-option, which defaults to s7c. <ul style="list-style-type: none"> <li>You can specify any command with or without its path.</li> </ul> </li> <li>Any detected error is shown in a "compilation" <a href="#">↗ Compilation Mode</a> buffer. Use it to navigate to the line of the code in error.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Emacs & Seed7 — References

| Document                       | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| The Seed7 Programming Language | <ul style="list-style-type: none"> <li><a href="#">Seed7 @ Wikipedia</a></li> <li><a href="#">Seed7 Home</a></li> <li><a href="#">Seed7 @ Github</a></li> </ul>                                                                                                                                                                                                                                                                                                                                   |  |
|                                | <ul style="list-style-type: none"> <li><a href="#">Seed7 @ reddit</a></li> <li><a href="#">Seed7 @ Rosetta code</a></li> <li><a href="#">Seed7-users mailing list archive</a></li> </ul>                                                                                                                                                                                                                                                                                                          |  |
| Articles on Seed7              | <ul style="list-style-type: none"> <li><a href="#">Modern Extensible Languages.</a> Daniel Zingaro, McMaster U. April 11, 2007 (pdf)</li> <li><a href="#">Meeting Seed7</a> - Norman Feske - Genome Labs, December 22, 2025</li> </ul>                                                                                                                                                                                                                                                            |  |
| Presentations                  | <ul style="list-style-type: none"> <li><a href="#">The Seed7 Programming Language @ Youtube</a></li> <li><a href="#">The Seed7 Programming Langage Presentation at CPP Vienna @ Youtube</a></li> <li><a href="#">Another speech about the Seed7 Programming Language</a></li> </ul>                                                                                                                                                                                                               |  |
| Emacs support                  | <a href="#">seed7-mode @ Github</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| Other tools that support Seed7 | <ul style="list-style-type: none"> <li><b>ripgrep</b> a very fast grep replacement - supports seed7 file types with <a href="#">this pull request accepted April 7 2025</a> <ul style="list-style-type: none"> <li>With this version of ripgrep, you can use <b>deadgrep</b> to identify Seed7 files by name in Emacs. See <a href="#">↗ Grep</a></li> </ul> </li> <li><b>ugrep</b> another very fast grep replacement - supports seed7 files with <a href="#">this pull request</a> .</li> </ul> |  |